Mainak Deb

Neuroscience PhD student @ Georgia Tech

(+91) 8335026861
mainakmayukh2000@gmail.com
mainakdeb.github.io
DOB: November 9th, 2000

Research

April 2026 **Model-derived Minimal Image Pairs Predict Differential Responses in Human Scene-Selective Cortex**
Junxia Wang, **Mainak Deb**, Haider Al-Tahan, Diego Garcia Cerdas, Iris Groen, Apurva Ratan Murty
Under review, CCN 2026

February 2025 **TopoNets: High performing vision and language models with brain-like topography**
Mayukh Deb, **Mainak Deb**, Apurva Ratan Murty.
ICLR spotlight 2025.

December 2024 **Small-scale adversarial perturbations expose differences between predictive encoding models of human fMRI responses**
Nikolas McNeal, **Mainak Deb**, Apurva Ratan Murty.
NeurIPS 2024 Unireps Workshop.

January 2024 **Microarchitectural Design Variation of the Echinoid Skeleton: A 3D Structural and Mechanical Study of Paracentrotud lividus**
Valentina Perricone, Pasquale Cesarano, **Mainak Deb**, Derek Lublin, Mirko Mutalipassi, Lucia Pappalardo, David Kisailus, Francesco Marmo.
(Preprint)

Internships and Industry

February 2025 **Research @ Tether Evo**
- Developed encoding models using pre-trained AI models to predict whole brain activity (fMRI) encoding multimodal input/stimuli (vision, audio and language). 15th Rank on algonauts 2025 challenge.

January 2024 **Research Intern @ MurtyLab (Dr. Ratan Murty), Georgia Tech**
- Developed projects dealing with adversarial robustness in models trained on brain data, inducing brain-like topography in ANNs, and simulating the effects of brain perturbations (lesion/stroke) on human visual perception.
- Leveraged fMRI datasets (NSD, DeepRecon, Murty185), built a python package (private) to seamlessly interface between fMRI data to pytorch dataloader.

July 2022 **Research Engineer @ Artflow.ai**
- Fine-tuning and building prompt systems to tame large language models like GPT-3 (davinci003), GPT-3.5-turbo and GPT-4.
- Leveraged language embeddings, audio embeddings and LLMs in tandem to facilitate automatic asset retrieval, including AI voices (replica and 11labs), music, shot-type, and sound sfx.
- Worked on 3D inpainting based camera movements and narrow depth-of-field effects on scene images to mimic the effect of high-end camera work.

July 2021 **Google Summer of Code 2021 Intern @ International Neuroinformatics Coordinating Facility**
Trained deep neural networks from scratch to help map the embryogenesis process in C. elegans worm embryo and deployed models live on the web using ONNX. Feel free to check out the detailed report and work repository

January 2022 **Research Intern @ Hybrid Design Lab, University of Campania, Italy**
Worked on generating time-varying textures that emulate Sea Urchin skeletons using Neural Cellular Automata and experimented with the latent output channels (apart from RGB) to generate vibrant psychedelic patterns. This was showcased at the Echino Design exhibition, 24 February 2022 at Città della Scienza, Naples, Italy.

March 2022 **Research Intern @ Department of Structures for Engineering and Architecture of University of Naples Federico II**
Worked with Prof. Francesco Marmo (and team) on bio-mechanical research that has the goal of geometrically and mechanically characterizing the stereom of echinoid exoskeletons. I processed micro-CT data to extract geometrical models of the stereom using computer vision, here's the Github repository and corresponding preprint

Personal Projects

December 2021 **Text-2-Neural Cellular Automata**
[Github, Blog post]
Generate beautiful cellular automata patterns from natural language prompts, using CLIP-guided Neural Cellular Automata, built using PyTorch.

December 2020 **Deceptive Digits**
[Github, Blog post]
Conditional GAN to generate hand written digit images.

November 2020 **Eyes on the Road**
Train a Pytorch CNN to classify driver activity (texting/talk on phone etc.), Test on my brother driving, for fun

August 2020 **Bank me Later**
Predict Bank Term deposit subscription probability using attributes like job, marital status, age etc with an accuracy of 94%

March 2020 **Deep Wine Connoisseur**
Use a pytorch dense net to predict wine-quality from it's chemical attributes.

February 2020 **Facial Expression Classifier**
PyTorch CNN to classify facial expressions w/ real-time inference

Hackathons

April 2021 **3rd Prize - MLOps for Good Hackathon**
Organized by Microsoft, Iguazio and MongoDB
Built and hosted Deepfake Shield - an online tool that uses deep-learning to detect deepfakes in an image.

April 2021 **First Prize (Education Track) - Hello World Hackathon**
Organized by CalHacks (University of California, Berkeley)
Built SignLingo - A deep learning based sign-language tutor which works via live webcam video feed.

October 2020 **First Prize - Lights, Camera, Hacktion! Hackathon**
Organized by Major League Hacking
Uses Computer Vision to automatically pause/play videos depending on user's attention to screen.

September 2020 **Third Prize - New Friends New Hacks**
Organized by Major League Hacking
Built an efficient Computer Vision based face mask detection system powered by OpenCV.

Education

2023 **Bachelor's of Technology**, *Electrical and Computer Engineering*, ASE, Amrita Vishwa Vidyapeetham, Kollam, India. GPA: 7.87/10.0

2016 - 2018 **Higher Secondary**, Amrita Vidyalayam, Kolkata , India. Marks: 89%

Technical Skills

Primary-Libraries PyTorch, OpenCV, NumPy, SciPy, Pandas, PIL, Matplotlib, Seaborn, Plotly, Slurm, Pycortex, Brainscore, Freesurfer